INCH-POUND

MIL-H-48590(AR)
AMENDMENT 2
17 June 1991
SUPERSEDING
AMENDMENT 1
30 April 1979

MILITARY SPECIFICATION

HEAD ASSEMBLY: 11727486
(PERISCOPE M32E1 and M35E1)

This amendment forms a part of Military Specification MIL-H-48590(MU), dated 31 December 1975, and is approved for use by the U.S. Army Armament, Munitions, and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

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3.6b, line 2, Delete "four" and substitute "three"

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3.9.1, Delete and substitute:

"3.9.1 Vibration "A". Vibration shall be applied along each of three mutually perpendicular axes of the head assembly. The vibration shall be sinusoidal as follows:

a. At a frequency of 5 Hertz (Hz), the acceleration shall be .5G ± 10%.

b. During a frequency sweep of 5 to 9 Hz, increase acceleration linearly from .5G to 1.5G's ± 10%.

c. During a frequency sweep of 9 to 500 Hz, maintain constant acceleration of 1.5 G's ± 10%.

d. During a frequency sweep of 500 to 9 Hz maintain constant acceleration of 1.5 G's ± 10%.

e. During a frequency sweep of 9 to 5 Hz, decrease acceleration linearly from 1.5 G's to .5 G ± 10%.

f. At 5 Hz, the acceleration shall be .5 G ± 10%.

The frequency of applied vibration shall be swept logarithmically in 15 minute ± 1 minute sweep cycles for a total of two hours per axis. The head assembly shall show no evidence of physical failure subsequent to vibration. Following the vibration test, the head assembly shall meet the requirements of 3.2, 3.5 through 3.8 inclusive and 3.10 through 3.13 inclusive."

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3.9.2, Delete and substitute:

"3.9.2 Vibration "B". Vibration shall be applied along each of three mutually perpendicular axes to the head assembly. The vibration shall be sinusoidal as follows:

a. At a frequency of 5 Hertz (Hz), the acceleration shall be 1.0 G ± 10%.

b. During a frequency sweep of 5 to 500 Hz, maintain constant acceleration of 1.0 G ± 10%.

c. During a frequency sweep of 500 to 5 Hz, maintain constant acceleration of 1.0 G ± 10%.

d. At 5 Hz, the acceleration shall be 1.0 G ± 10%.

The frequency of applied vibration shall be swept logarithmically in 15 minute ± 1 minute sweep cycles for a total of 15 minutes ± 1 minute along each of the three mutually perpendicular axes (total vibration time 45 minutes ± 3 minutes). The head assembly shall show no evidence of physical failure subsequent to vibration. Following the vibration test, the head assembly shall meet the requirements of 3.2, 3.5 through 3.8 inclusive and 3.10 through 3.13 inclusive."

3.9.3, Delete and substitute:

"3.9.3 Shock. The head assembly shall be subjected to 24 cumulative shocks consisting of three each at 35 G's, ± 10%, then 20 G's ± 10% applied in each direction along each of the two mutually perpendicular horizontal axes (all four sides). The shock pulse shall have a half sine wave with a time duration of 3 ± .2 milliseconds. The head assembly shall also be subjected to six cumulative shocks consisting of three each at 35 G's ± 10%, then 20 G's ± 10% applied in one direction along the verticle axis (normal operating position of head assembly only). The shock pulse shall have a half sine wave with a time duration of 18 ± 3 milli-seconds. The head assembly shall show no evidence of physical failure subsequent to shock. Following the shock test, the head assembly shall meet the requirements of 3.2, 3.5 through 3.8 inclusive and 3.10 through 3.13 inclusive."

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* 3.12, Delete and substitute:

"3.12 Sealing. The head assembly shall show no evidence of leakage in excess of 0.1 pound per square inch (psi) when subjected to an internal pressure of 5. ± .010 psi for a minimum of 1 hour. The interior atmosphere of the accepted head assembly shall contain nitrogen (having a dew point at least as low as minus 25°F at atmospheric pressure) between 0.25 and 0.50 pound per square inch gage (psig)."

3.13, line 3, Delete "0.05" and substitute "0.05 inch"

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* 4.6.10, line 2, Delete "0.25 to 0.50" ad substitute "0.0 to 0.50"

NOTE: The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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